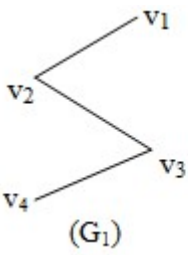
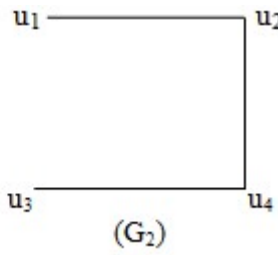


Course Code:B20BS1104																					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)			R 20																		
I B. Tech I Semester –MODEL PAPER																					
DISCRETE MATHEMATICS																					
(For CSBS)																					
Time: 3 Hrs.			Max. Marks:70																		
Answer any one Question from Each Unit																					
All questions carry equal Marks																					
			M	CO	KL																
UNIT - I																					
1) (a)	Using the truth table establish that $[\sim q \wedge (p \rightarrow q)] \rightarrow (\sim p)$ is a tautology.			7	CO1	K3															
(b)	Determine the conjunctive normal form of $p \wedge (p \rightarrow q)$.			7	CO1	K3															
(OR)																					
2) (a)	Establish that $\sim(p \wedge q) \rightarrow [\sim p \vee (\sim p \vee q)] \Leftrightarrow (\sim p \vee q)$.			7	CO1	K3															
(b)	Determine the principle disjunctive normal form and principle conjunctive normal form for $p \vee (\neg p \rightarrow (q \vee (\neg q \rightarrow r)))$.			7	CO1	K3															
UNIT - II																					
3) (a)	Establish that, if 'n' is a positive integer and p^2/n , where p is a prime number, then D_n is not a Boolean algebra.			7	CO2	K3															
(b)	Establish that in a Boolean algebra, for any 'a' and 'b', $a \leq b$ iff $(a * b) \oplus (a' * b) = 0$.			7	CO2	K3															
(OR)																					
4) (a)	Consider the Boolean polynomial $p(x, y, z) = x * (y \oplus z')$. If $B=\{0,1\}$, compute the truth table of the function $f: B_3 \rightarrow B$ defined by p . Also draw logical diagram.			7	CO2	K3															
(b)	Develop the K-map for the function whose truth table is given by <table border="1"><tr><td>x</td><td>y</td><td>f(x,y)</td></tr><tr><td>0</td><td>0</td><td>1</td></tr><tr><td>0</td><td>1</td><td>0</td></tr><tr><td>1</td><td>0</td><td>0</td></tr><tr><td>1</td><td>1</td><td>1</td></tr></table> Utilize this K-map to determine the function $f(x, y)$.			x	y	f(x,y)	0	0	1	0	1	0	1	0	0	1	1	1	7	CO2	K3
x	y	f(x,y)																			
0	0	1																			
0	1	0																			
1	0	0																			
1	1	1																			
UNIT - III																					
5) (a)	Suppose R is a relation on a set of all ordered pairs of positive integers defined as $(x, y)R(u, v)$ iff $xv = uv$. Establish that R is an equivalence relation.			7	CO3	K3															
(b)	Let 'E' be the set of all even integers.Establish that the semi groups $(Z,+)$ and $(E,+)$ are isomorphic.			7	CO3	K3															
(OR)																					
6) (a)	Let 'R' be the set of real numbers and * denote the binary operation defined by $a*b=a+b+3ab$. Then establish the fact that $(R, *)$ is a semigroup. Is it a monoid? If so specify the identity element.			7	CO4	K3															
(b)	Establish that the set of all non-zero complex numbers is a group under the			7	CO4	K3															

	operation of multiplication.			
UNIT - IV				
7) (a)	Determine the generating function for the sequence $1^2, 2^2, 3^2 \dots$	7	CO5	K3
(b)	Solve the recurrence relation $a_n = a_{n-1} + a_{n-2}$ with $a_0 = 0, a_1 = 1$.	7	CO5	K3
(OR)				
8) (a)	Utilize mathematical induction to prove that $3 + (3x5) + (3x5^2) + \dots + (3x5^n) = \frac{3(5^{n+1}-1)}{4}$.	7	CO5	K3
(b)	Determine the number of positive integers between 1 and 250 inclusive which are divisible by 3, 5 or 7.	7	CO5	K3
UNIT - V				
9) (a)	Establish an isomorphism of the following graphs  (G_1)  (G_2)	7	CO6	K3
(b)	If a connected planar graph 'G' has n_v vertices, n_e edges and n_f regions then establish that $n_v - n_e + n_f = 2$.	7	CO6	K3
(OR)				
10) (a)	Establish the fact that a tree with n vertices has n-1 edges.	7	CO6	K3
(b)	Suppose that a tree 'T' has 'N' vertices of degree '1', 2 vertices of degree '2', 4 vertices of degree '3' and 3 vertices of degree '4', determine 'N'.	7	CO6	K3

Course Code: B20BS1105												
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)									R 20			
I B. Tech I Semester –MODEL PAPER												
INTRODUCTORY TOPICS IN STATISTICS, PROBABILITY AND CALCULUS												
(For CSBS)												
Time: 3 Hrs.									Max. Marks:70			
Answer any one Question from Each Unit												
All questions carry equal Marks												
Q.No	UNIT - I									M	CO	KL
1.a)	Explain the methods related to primary and secondary data.									7	CO1	K2
b)	Determine median and quartiles for the following data.									7	CO1	K3
	Marks	0-10	10-20	20-30	30-40	40-50	50-60	60-70	70-80			
	No. of students	11	18	25	28	30	33	22	15			
OR												
2.a)	Write short notes on Histogram and Ogive curves.									7	CO1	K2
b)	An analysis of monthly wages paid to workers of two firms A and B belonging to the same industry gives the following result:									7	CO1	K3
			Firm-A		Firm-B							
	Number of workers		500		600							
	Average daily wage		Rs. 186.00		Rs. 175.00							
	Variance of wages		81		100							
	(1) Which firm, A or B has a larger wage bill?											
(2) In which firm, A or B, is there greater variability in individual wages?												
(3) Calculate the average and variance of the wages of all workers in the firms A and B taken together.												
UNIT – II												
3.a)	State and prove Bayes Theorem.									7	CO2	K3
b)	A committee of 4 people is to be appointed from 3 officers of the production department, 4 officers of the purchase department, 2 officers of the sales department and 1 chartered accountant. Determine the probability of forming the committee in the following manner:									7	CO2	K3
	(i) There must be one from each category.											
	(ii)It should have at least one from the purchase department											
OR												
4. a)	Data on the readership of a certain magazine show that the proportion of male readers under 35 is 0.40 and over 35 is 0.20. if the proportion of readers under 35 is 0.70, determine the proportion of subscribers that are females over 35 years. Also determine the probability for a randomly selected male subscriber to be under 35 years age.									7	CO2	K3

b)	The probabilities of X,Y and Z becoming managers are 4/9, 2/9 and 1/3 respectively. The probabilities that the Bonus scheme will be introduced if X,Y and Z becomes managers are 3/10, 1/2 and 4/5 respectively. (i) Determine the probability for the Bonus scheme to be introduced? (ii) If the Bonus scheme has been introduced, determine the probability for X to be the appointed manager?	7	CO2	K3
UNIT - III				
5. a)	Establish that mean and variance of a poison distribution are equal.	7	CO3	K3
b)	A random variable X has the probability density function given by $f(x) = \begin{cases} x & \text{if } 0 \leq x < 1 \\ 2 - x & \text{if } 1 \leq x < 2 \\ 0 & \text{elsewhere} \end{cases}$ Determine E(X) and V(X).	7	CO3	K3
OR				
6. a)	Ten coins are thrown simultaneously. Find the probability of getting at least i)seven heads ii) six heads	7	CO4	K3
b)	Define moment generating function and write its properties.	7	CO4	K2
UNIT - IV				
7.a)	Determine the mean and variance of Exponential distribution.	7	CO5	K3
b)	In a test on 2000 electric bulbs, it was found that the life of a particular make, was normally distributed with an average life of 2040 hours and S.D of 60 hours. Estimate the number of bulbs likely to burn for (i) more than 2150 hours (ii) more than 1920 hours and less than 2160 hours.	7	CO5	K3
OR				
8. a)	Write short notes on Chi-square distribution.	7	CO5	K2
b)	In a normal distribution 31% of the items are under 45 and 8% are over 64. Determine mean and standard deviation of the distribution.	7	CO5	K3
UNIT-V				
9. a)	If $U = \sin^{-1} \left(\frac{x+y}{\sqrt{x}+\sqrt{y}} \right)$, establish that $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = \frac{1}{2} \tan u$	7	CO6	K3
b)	Determine the maximum and minimum values of $x^3 + y^3 - 3x^2y$.	7	CO6	K3
OR				
10. a)	Evaluate $\iint x y (x + y) dy dx$ over the area bounded by $y = x^2$ and $y = x$.	7	CO6	K3
b)	Determine the area of the circle $x^2 + y^2 = a^2$ using double integrals.	7	CO6	K3

Course Code: B20BS1106					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)				R 20	
I B. Tech I Semester –MODEL PAPER					
FUNDAMENTALS OF PHYSICS					
(For CSBS)					
Time: 3 Hrs.			Max. Marks:70		
Answer any one Question from Each Unit					
All questions carry equal Marks					
			M	CO	KL
UNIT-I					
1	a.	Solve the differential equation of a damped harmonic oscillator. Investigate the conditions of light damping, heavy damping and critical damping	14	1	3
OR					
2	a.	What is simple harmonic motion? Explain its characteristics	8	1	2
	b.	Explain the energy decay in a damped harmonic oscillator	6	1	3
UNIT-II					
3	a.	Explain the principle of Superposition and discuss the conditions for maxima and minima of intensity.	6	2	2
	b.	How the Newton's Rings are formed and deduce an expression for the wave length of light used.	8	2	3
OR					
4	a.	Distnguish between Interference and Diffraction of light.	4	2	1
	b.	Explain the diffraction phenomena in grating and deduce the grating equation	5	2	3
	c.	What are the different types of polarized lights and how they are obtained?	5	2	2
UNIT-III					
5	a.	What is a wave function and explain its significance.	6	3	2
	b.	Deduce the eigen values and eigen functions associated with particle in one dimensional box using the Schrodinger time independent wave equation.	8	3	3
OR					
6	a.	What is a Bravais Lattice and explain the Bravais lattice in different Crystal systems.	8	3	3
	b.	Give classifications of the materials using band theory of solids with the help of suitable diagrams	6	3	2
UNIT-IV					
7	a.	Give the selection procedure of the active medium of laser device.	6	4	2
	b.	With suitable diagrams, discuss the working principle, design and working of Ruby laser system	8	4	3
OR					
8	a.	Explain characteristics of Laser in detail	6	4	2
	b.	Explain the principle of light propagation in optical fiber, discuss the sensor applications of optical fiber	8	5	3
UNIT-V					
9	a.	State and explain First law of thermodynamics.	6	6	2
	b.	Deduce an expression for the efficiency of a Carnot engine describing various cycles of operation.	8	6	3

OR					
10	a.	Explain the concept of entropy.	10	6	2
	b.	The efficiency of Carnot's ideal heat engine has increased from 40 % to 50 % if the source temperature is raised by 100°C . Determine the temperature of source and sink ?	4	6	2

Course Code: B20EE1103

SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)

R 20

I B. Tech I Semester MODEL QUESTION PAPER

PRINCIPLES OF ELECTRICAL ENGINEERING

(for CSBS)

Time: 3 Hrs.

Max. Marks: 70

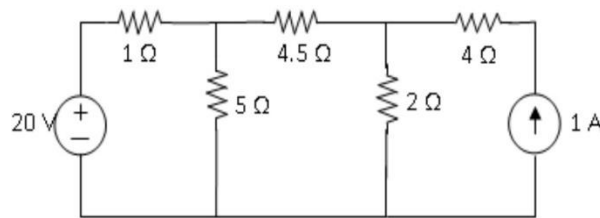
Answer any one Question from Each Unit

All questions carry equal Marks

M CO KL

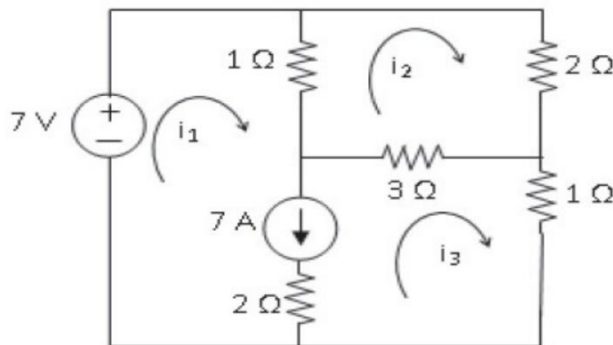
UNIT-I

1	a.	Explain about different types of sources.	7M	1	2
	b.	Use Nodal Analysis to find currents in different resistances of the circuit shown.	7M	1	3



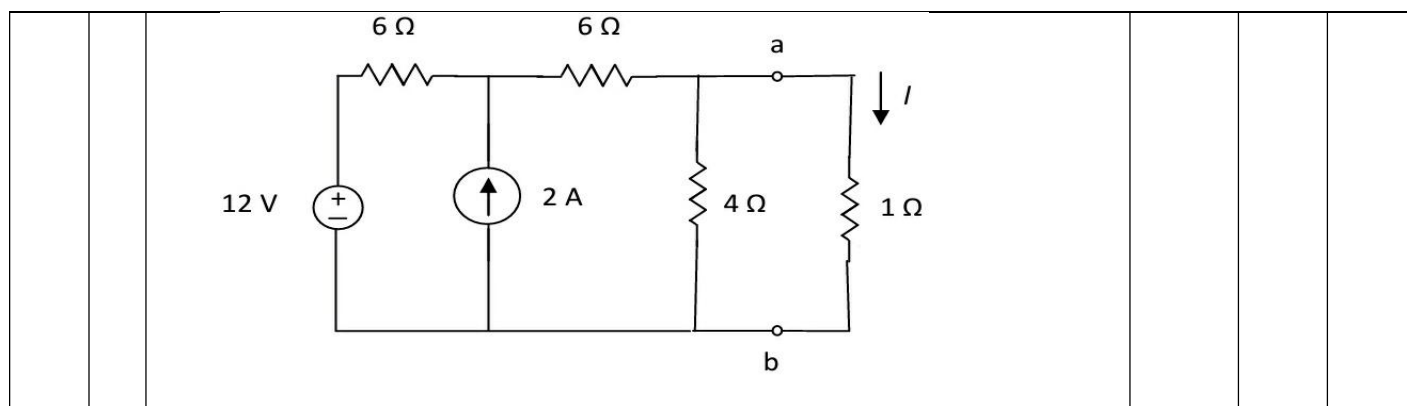
OR

2	a.	State and explain Kirchhoff's laws with examples.	7M	1	3
	b.	Find the current in 3ohm resistor and voltage across the current source in the network using Mesh Analysis.	7M	1	3



UNIT-II

3	a.	Derive the relations for star- delta transformations.	7M	2	3
	b.	Determine the current I through the 1 ohm resistance in the circuit shown using Thevenin's theorem.	7M	2	3



OR

4	a.	Determine the current through 10 ohm resistance in the network shown using star- delta transformation.	7M	2	3
	b.	Find the current I in the circuit shown using superposition theorem.	7M	2	3

UNIT-III

5	a.	Discuss the analysis of single phase ac circuit consisting of series RL.	7M	3	3
	b.	A 200V, 50 Hz AC supply is applied to a coil of 0.08H inductance and 3.5 ohm resistance connected in series with a 0.0072mF capacitor. Calculate (i) Impedance (ii) Current (iii) phase angle between current and voltage (iv) power factor (v) power consumed.	7M	3	3
OR					
6	a.	Explain (i) Active power (ii) Reactive power (iii) Apparent power (iv) Complex power in ac circuits.	7M	3	2

	b.	Obtain relation between line and phase quantities in a three phase star connected circuit.	7M	3	3
UNIT-IV					
7	a.	Explain about charging and discharging of a capacitor.	7M	4	3
	b.	State and explain Faraday's laws of electromagnetic induction.	7M	4	3
OR					
8	a.	Derive the EMF equation of a single phase transformer.	7M	4	3
	b.	An ideal 25KVA transformer has 500 turns on primary and 400 turns on secondary winding. The primary winding is connected to 3000V, 50 Hz supply. Calculate (i) primary and secondary currents. (ii) secondary EMF. (iii) maximum flux.	7M	4	3
UNIT-V					
9	a.	Explain the methods for the measurement of single phase power.	7M	5	3
	b.	Give the basic layout of distribution system.	7M	5	2
OR					
10	a.	What is the necessity of Earthing? Explain different types of Earthing.	7M	5	2
	b.	Explain about different types of safety devices.	7M	5	2

Course Code: B20CB1101					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R 20
I B. Tech I Semester –MODEL PAPER					
FUNDAMENTALS OF COMPUTER SCIENCE & PROGRAMMING					
(For CSBS)					
Time: 3 Hrs.			Max. Marks:70		
Answer any one Question from Each Unit					
All questions carry equal Marks					
			M	CO	KL
UNIT - I					
1.	a).	Describe various types of computer languages	7M	CO1	K2
	b).	Write about different types of notations in flow chart?	7M	CO1	K2
OR					
2.	a).	Explain various data types in C Language	7M	CO1	K2
	b).	Define an operator? Explain various types of Bitwise Operators with examples.	7M	CO1	K2
UNIT - II					
3.	a).	Differentiate between if statement and if-else statement with suitable examples and proper syntax	7M	CO2	K3
	b).	Define a function? Explain various categories of functions	7M	CO2	K2
OR					
4.	a).	Write a program whether given number is Armstrong number or not	7M	CO2	K3
	b).	Define a recursive function? Write a program for GCD of two numbers using recursion	7M	CO2	K3
UNIT - III					
5.	a).	How to declare, initialize and reading of one dimensional array with an example	7M	CO3	K2
	b).	Define a pointer? explain pointer arithmetic	7M	CO3	K1
OR					
6.	a).	Write a C program for multiplication of two matrices	7M	CO3	K3
	b).	Differentiate between Call-by-Value and Call-by-Reference with an example?	7M	CO3	K3
UNIT - IV					
7.	a).	Define a structure? How to declare, initialize, accessing of structure elements with example.	7M	CO4	K2
	b).	Explain Dynamic Memory Allocation	7M	CO4	K2
OR					
8.	a).	Explain pointer of Structures with an example	7M	CO4	K2
	b).	Outline C Pre-processor and write various pre-processor commands	7M	CO4	K2
UNIT - V					
9.	a).	Explain various types of file functions?	7M	CO5	K2
	b).	Write about Unix system calls (read and write, open, create, close)?	7M	CO5	K2
OR					
10.	a).	Write a program to read text into the file and print the text from file?	7M	CO5	K3
	b).	Explain about Macro and User defined Libraries.	7M	CO5	K2

Course Code: B20BS1205																					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R 20																
I B. Tech II Semester –MODEL PAPER																					
LINEAR ALGEBRA																					
(For CSBS)																					
Time: 3 Hrs.			Max. Marks:70																		
Answer any one Question from Each Unit																					
All questions carry equal Marks																					
Q. No.		UNIT-1	M	CO	KL																
1	A	Evaluate the Determinant of $A = \begin{bmatrix} 0 & 4 & -1 & 5 \\ -4 & 0 & 3 & -2 \\ 1 & -3 & 0 & 1 \\ -5 & 2 & -1 & 0 \end{bmatrix}$, by triangle method.	7	CO -1	K3																
	B	Solve the system of equations by Cramer's rule: $x-3y-4z = 1$; $-x + y - 3z = 14$; $y - 3z = 5$.	7	CO -1	K3																
OR																					
2	A	Solve the system of equations by Gauss elimination method: $3x-y-2z = 1$; $x - 2y - 4z = 16$; $3x - y + 4z = 15$.	7	CO -1	K3																
	B	Determine the Inverse of $A = \begin{bmatrix} -1 & 1 & 2 \\ 3 & -1 & 1 \\ -1 & 3 & 4 \end{bmatrix}$ by using Gauss Jordan method	7	CO -1	K3																
UNIT-2																					
3	A	Determine whether the vector (2,1,3) is a linear combination of the vectors (1,2,3) and (2,3,1).	7	CO -2	K3																
	B	Determine the rank of the matrix: $A = \begin{bmatrix} 15 & 13 & 2 & 2 \\ 12 & 5 & 8 & 4 \\ -11 & 3 & 20 & -3 \\ 7 & -5 & 8 & 1 \end{bmatrix}$, by using Gauss elimination.	7	CO -2	K3																
OR																					
4	A	An electrical engineer supervises the production of three types of electrical components. Three kinds of material - metal, plastic, and rubber are required for production. The amounts needed to produce each component are <table border="1"><thead><tr><th>compo nent</th><th>Metal (g/com)</th><th>Plastic (g/com)</th><th>Rubber (g/com)</th></tr></thead><tbody><tr><td>1</td><td>15</td><td>0.30</td><td>1.0</td></tr><tr><td>2</td><td>17</td><td>0.40</td><td>1.2</td></tr><tr><td>3</td><td>19</td><td>0.55</td><td>1.5</td></tr></tbody></table> If totals of 3.89, 0.095 and 0.282 kg of metal, plastic and rubber respectively are available each day, determine the number of components that can be produced per day, by applying LU decomposition method?	compo nent	Metal (g/com)	Plastic (g/com)	Rubber (g/com)	1	15	0.30	1.0	2	17	0.40	1.2	3	19	0.55	1.5	7	CO -2	K3
	compo nent	Metal (g/com)	Plastic (g/com)	Rubber (g/com)																	
1	15	0.30	1.0																		
2	17	0.40	1.2																		
3	19	0.55	1.5																		
B	Solve the equations $9x+2y+4z=20$; $x+10y+4z=6$; $2x-4y+10z=-15$ by Gauss-Seidel method.		7	CO -2	K3																
UNIT-3																					
5	A	Establish that the set $S = \{(1,2,1), (2, 1, 0), (1, -1, 2)\}$ forms a basis for $V_3(F)$	7	CO -3	K3																
	B	Let W be the plane generated by the vectors (1,1,1) and (1,0,1). Determine the orthogonal projection $P:R^3 \rightarrow W$	7	CO -3	K3																

OR					
6	A	Apply Gram-Schmidt orthogonalization to the following sequence of vectors in $\mathbb{R}^3$ $= \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}; \begin{bmatrix} 8 \\ 1 \\ -6 \end{bmatrix}; \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$ and determine orthogonal basis and orthonormal basis	7	CO -4	K3
	B	Determine the QR decomposition of A, where $A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 0 \\ -1 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix}$	7	CO -4	K3
UNIT-4					
7	A	Determine the Eigen values and Eigen vectors of the matrix $A = \begin{bmatrix} 3 & -6 & 3 \\ 1 & 0 & -1 \\ 1 & 2 & -3 \end{bmatrix}$	7	CO -5	K3
	B	If $T: V_3(\mathbb{R}) \rightarrow V_2(\mathbb{R})$ defined by $T(a_1, a_2, a_3) = (a_1 - a_2, a_2 - a_1)$ Establish that T is Linear Transformation.	7	CO -5	K3
OR					
8	A	Determine the Range, Rank, Null space of $T: \mathbb{R}^2 \rightarrow \mathbb{R}^3$ defined by $T(x_1, x_2) = (x_1 + x_2, x_1 - x_2, x_2)$.	7	CO -5	K3
	B	Determine whether the matrix $A = \begin{bmatrix} 2 & 3-5i & 1+i \\ 3+5i & -1 & 2-i \\ 1-i & 2+i & 3 \end{bmatrix}$ is a skew-Hermitian matrix or not.	7	CO -5	K3
UNIT-5					
9	A	Establish that the squared error of SVD at a truncation rank of (k+1) is no larger than the squared error of PCA at a truncation rank of k for any $k \geq 1$.	7	CO -6	K3
	B	Determine the singular values of A: $A = \begin{bmatrix} 0 & 1 & 1 \\ \sqrt{2} & 2 & 0 \\ 0 & 1 & 1 \end{bmatrix}$ and Determine the SVD decomposition of A.	7	CO -6	K3
OR					
10	A	If these flags have rank 2, establish A and B in any way as $u_1 v_1^T + u_2 v_2^T$. $A_{\text{Sweden}} = A_{\text{Finland}} = \begin{bmatrix} 1 & 2 & 1 & 1 \\ 2 & 2 & 2 & 2 \\ 1 & 2 & 1 & 1 \end{bmatrix}; B_{\text{Benin}} = \begin{bmatrix} 1 & 2 & 2 \\ 1 & 3 & 3 \end{bmatrix}$	7	CO -6	K3
	B	Singular Value Decomposition of matrix A is given by the following : $A = \begin{bmatrix} -1/\sqrt{2} & 1/\sqrt{2} \\ 1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix} \begin{bmatrix} 4 & 0 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}^T$. Determine the inverse of A without explicitly materializing A.	7	CO -6	K3

Course Code: B20BS1206																																															
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)											R 20																																				
I B. Tech II Semester –MODEL PAPER																																															
STATISTICAL METHODS																																															
(For CSBS)																																															
Time: 3 Hrs.											Max. Marks:70																																				
Answer any one Question from Each Unit																																															
All questions carry equal Marks																																															
Q.No	UNIT - I											M	CO	KL																																	
1.a)	The following results were obtained from records of age (x) and blood pressure (y) of a group of 10 persons. Mean($\bar{x}$) = 53 ;Mean($\bar{y}$) = 142 Variance(x) = 142; Variance (y) = 165 ; $\sum(x - \bar{x})(y - \bar{y}) = 1220$. Determine the appropriate regression line and use it to estimate the blood pressure (y) Of a person whose age is 45.											7	CO1	K3																																	
b)	Write short notes on (i) Scatter diagram (ii) Rank Correlation Coefficient.											7	CO1	K1																																	
(OR)																																															
2.a)	Ten participants in a contest are ranked by two judges as follows <table border="1"><tr><td>Serial number</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td><td>8</td><td>9</td><td>10</td></tr><tr><td>x</td><td>1</td><td>6</td><td>5</td><td>10</td><td>3</td><td>2</td><td>4</td><td>9</td><td>7</td><td>8</td></tr><tr><td>y</td><td>6</td><td>4</td><td>9</td><td>8</td><td>1</td><td>2</td><td>3</td><td>10</td><td>5</td><td>7</td></tr></table> Determine the rank correlation coefficient.											Serial number	1	2	3	4	5	6	7	8	9	10	x	1	6	5	10	3	2	4	9	7	8	y	6	4	9	8	1	2	3	10	5	7	7	CO1	K3
Serial number	1	2	3	4	5	6	7	8	9	10																																					
x	1	6	5	10	3	2	4	9	7	8																																					
y	6	4	9	8	1	2	3	10	5	7																																					
b)	Fit a linear regression of rice yield (X_1 quintals) on the use of fertilizer(X_2 kgs per acre) and the amount of rainfall X_3 inches), from the following data. Estimate the yield when $X_2 = 60$ and $X_3 =25$. <table border="1"><tr><td>X_1</td><td>45</td><td>50</td><td>55</td><td>70</td><td>75</td><td>75</td><td>85</td></tr><tr><td>X_2</td><td>25</td><td>35</td><td>45</td><td>55</td><td>65</td><td>75</td><td>85</td></tr><tr><td>X_3</td><td>31</td><td>28</td><td>32</td><td>32</td><td>29</td><td>27</td><td>31</td></tr></table>											X_1	45	50	55	70	75	75	85	X_2	25	35	45	55	65	75	85	X_3	31	28	32	32	29	27	31	7	CO1	K3									
X_1	45	50	55	70	75	75	85																																								
X_2	25	35	45	55	65	75	85																																								
X_3	31	28	32	32	29	27	31																																								
UNIT – II																																															
3.a)	Explain (i) Types of sampling (ii) Sampling distribution											7	CO2	K2																																	
b)	Suppose the diameter of motor shafts in a lot have a mean of 0.249 inches and s.d. of 0.003inches. The inner diameter of bearings in another lot have a mean of 0.255inches and s.d. of 0.002 inches. Determine the mean and s.d. of clearness between shafts and bearings selected from these lots.											7	CO2	K3																																	
(OR)																																															
4.	A population consists of four numbers 2,3,4 and 5. Consider all possible distinct samples of size two with replacement. Determine (a) the population mean (b) the population standard deviation (c) the sampling distribution of means (d) the mean of the s.d of means (e) s.d. of sampling distribution of standard deviation.											14	CO2	K3																																	
UNIT - III																																															
5. a)	Fit a straight line to the following data by method of least squares <table border="1"><tr><td>x</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td></tr><tr><td>y</td><td>1</td><td>1.8</td><td>1.3</td><td>2.5</td><td>6.3</td></tr></table>											x	0	1	2	3	4	y	1	1.8	1.3	2.5	6.3	7	C03	K3																					
x	0	1	2	3	4																																										
y	1	1.8	1.3	2.5	6.3																																										

b)	Explain fitting of a parabola for a data and determine the normal equations.	7	C03	K3																		
(OR)																						
6. a)	Describe (i) Point estimation (ii) Interval estimation.	7	CO4	K3																		
b)	Determine the maximum likelihood estimate for the parameter λ of the Poisson distribution on the basis of sample size ‘n’.	7	CO4	K3																		
UNIT - IV																						
7.a)	Explain the method of ‘Analysis of Variance one-way classification’.	7	C05	K2																		
b)	The manager of Akash software company wishes to study the number of hours senior executives by types of industry spend at their desktop computers. The manager selected a sample of 5 executives from each of three industries- Banking, Retail and Insurance. Test at 5% level of significance whether there is a difference in the average number of hours spent per week by industry. <table><tr><td>Banking</td><td>12</td><td>10</td><td>11</td><td>12</td><td>10</td></tr><tr><td>Retail</td><td>8</td><td>8</td><td>6</td><td>8</td><td>10</td></tr><tr><td>Insurance</td><td>10</td><td>6</td><td>6</td><td>8</td><td>8</td></tr></table>	Banking	12	10	11	12	10	Retail	8	8	6	8	10	Insurance	10	6	6	8	8	7	CO5	K3
Banking	12	10	11	12	10																	
Retail	8	8	6	8	10																	
Insurance	10	6	6	8	8																	
(OR)																						
8.	Explain any three forecasting techniques in Time series analysis.	14	C05	K2																		
UNIT-V																						
9. a)	Explain the steps in Testing of hypothesis. Also define Type I error and Type II error.	7	CO6	K2																		
b)	Following are the responses to the question “How many hours do you study before a major mathematics test?” 6 5 1 2 2 5 7 5 3 7 4 7 Use the Wilcoxon signed-rank test to test the hypothesis at the 0.05 level of significance that the median number of hours of study before a major mathematics test is 3 hours.	7	CO6	K3																		
(OR)																						
10. a)	Describe Mann – Whitney test.	7	CO6	K3																		
b)	A coin is tossed 20 times and the following sequence of heads (H) and tails (T) is obtained. H T T H H T T H T H H H T T H H T H H H Use Run test to determine at 5% level of significance if the coin is unbiased.	7	CO6	K3																		

Course Code: B20CB1201					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R 20
I B. Tech II Semester –MODEL PAPER					
DATA STRUCTURES AND ALGORITHMS					
(For CSBS)					
Time: 3 Hrs.			Max. Marks:70		
Answer any one Question from Each Unit					
All questions carry equal Marks					
		UNIT - I	M	CO	KL
1.	a).	Explain about Asymptotic Notations with Examples.	7M	CO1	K2
	b).	Evaluate the time complexity of Towers of Hanoi problem?	7M	CO1	K2
		OR			
2.	a).	Define an Algorithm? Explain various types of Algorithms?	7M	CO1	K2
	b).	Define Data Abstraction? Explain Time-Space Off?	7M	CO1	K2
		UNIT - II			
3.	a).	Write C routines for the operations of single Linked list.	7M	CO2	K3
	b).	Define Queue. Explain the operations of queue using arrays.	7M	CO2	K2
		OR			
4.	a).	Explain polynomial addition using Linked List.	7M	CO2	K2
	b).	Write about application of Linear Data Structures.	7M	CO2	K2
		UNIT - III			
5.	a).	What is a Binary Tree? Explain threaded binary tree	7M	CO3	K2
	b).	Define Binary Search Tree. Explain about insertion and deletion of an element from Binary Search Tree	7M	CO3	K2
		OR			
6.	a).	Define AVL Tree. Write algorithms for insertion and deletion of elements from AVL Tree.	7M	CO3	K3
	b).	Explain about B+ Tree.	7M	CO3	K2
		UNIT - IV			
7.	a).	Write breadth first Search traversal algorithm. Explain with an example.	7M	CO4	K2
	b).	Write an algorithm for binary search using iterative and Explain with Example	7M	CO4	K2
		OR			
8.	a).	Write an algorithm for Quick Sort and Rearrange following numbers using quick sort: 10, 6, 3, 7, 17, 26, 56, 32, 72	7M	CO4	K4
	b).	Rearrange following numbers using Heap Sort: 24, 12, 56, 43, 87, 65, 22,21	7M	CO4	K4
		UNIT - V			
9.	a).	What is Hashing? Write about Linear Probing in Hashing?	7M	CO5	K2
	b).	Write about Indexed Sequential accessing scheme?	7M	CO5	K2
		OR			
10.	a).	Write an Algorithm to find single source shortest path in a Graph with Example	7M	CO5	K3
	b).	Explain about Interpolation Search.	7M	CO5	K2

Course Code: B20EC1201					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R 20
I B. Tech II Semester –MODEL PAPER					
PRINCIPLES OF ELECTRONICS ENGINEERING					
(For CSBS)					
Time: 3 Hrs.			Max. Marks:70		
Answer any one Question from Each Unit					
All questions carry equal Marks					
		UNIT-I	CO	KL	M
1.	a).	Explain how materials are classified based on energy gap?	1	K2	6
	b).	Write about intrinsic and extrinsic (both p-type and n-type) semiconductors?	1	K2	8
		OR			
2.	a).	Explain how the conductivity of semiconductors depends on temperature?	1	K2	6
	b).	Explain Drift and Diffusion phenomenons in semiconductors?	1	K2	8
		UNIT-II			
3.	a).	Explain operation and V-I characteristics of p-n junction diode in forward & reverse bias conditions?	1	K2	8
	b).	Compare Avalanche and Zener breakdown phenomenons?	1	K2	6
		OR			
4.	a).	Explain the operation of fullwave rectifier with a neat sketch?	1	K2	7
	b).	Describe Zener breakdown? Explain the operation of Zener diode as a voltage regulator?	1	K2	7
		UNIT-III			
5.	a).	Explain operation and characteristics of transistor in CB cofiguration with a neat sketch?	2	K2	8
	b).	Compare transistor CB, CE and CC configurations?	2	K3	6
		OR			
6.	a).	Explain construction and operation of p- channel JFET with a neat sketch ?	2	K2	8
	b).	Compare BJT and JFET?	2	K3	6
		UNIT-IV			
7.	a).	Explain the concept of feed-back with a neat block diagram?	3	K2	8
	b).	Write note on op-amp as i) adder ii) integretror iii) differentiator.	3	K1	6
		OR			
8.	a).	Write a note on topologies of feedback amplifier?	3	K2	7
	b).	Explain BJT RC phase shift oscillator with a neat sketch?	3	K1	7
		UNIT-V			
9.	a).	Perform the following conversions: (i) $(233)_{10} = ()_8$ (ii) $(11111011)_2 = ()_{16}$ (iii) $(ABC)_{16} = ()_8$ (iv) $(111.011)_2 = ()_{10}$	4	K3	8
	b).	Write a short note on i)NAND ii) NOR iii) EX-OR gates with thier relevent logical symbols and truth tables.	4	K1	6
		OR			
10	a).	Write a short note on i) Full adder using two-half adders ii) Ripple counters	4	K1	6
	b).	Write a short note on i) SR flip-flop ii) SIPO shift-register iii) 4x1 Demultiflexer	4	K1	8

Course Code: B20HS1201						
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)				R 20		
I B. Tech II Semester –MODEL PAPER						
FUNDAMENTALS OF ECONOMICS						
(For CSBS)						
Time: 3 Hrs.			Max. Marks:70			
Answer any one Question from Each Unit						
All questions carry equal Marks						
UNIT - I				M	KL	CO
1.	a).	Define Economics and explain about Micro Economics?	7	2	1	
	b).	Explain the types of Elasticity of Demand?	7	3	2	
OR						
2.	a).	What is Demand Forecasting and Explain about Survey and Opinion methods?	7	3	2	
	b).	Define Supply and Explain about the Law of Supply?	7	3	2	
UNIT – II						
3.	a).	Write the difference between Positive Economics and Welfare Economics?	7	2	3	
	b).	Explain the Law of Diminishing Marginal Utility and its assumptions?	7	2	3	
OR						
4.	a).	Define Indifference Technique and Explain the assumptions of Indifference Curve?	7	2	3	
	b).	Draw an Indifference curve, with the help of the schedule?	7	2	3	
UNIT – III						
5.	a).	Define Production and Explain the Production function?	7	2	4	
	b).	Write about various Cost concepts?	7	2	4	
OR						
6.	a).	Explain Cost-Output relationship in Short Run and Long Run?	7	2	4	
	b).	Write about the features of Monopoly and Oligopoly?	7	2	4	
UNIT – IV						
7.	a).	Define National Income and Explain GNP, NNP?	7	2	5	
	b).	Write the methods for measuring National Income?	7	2	5	
OR						
8.	a).	Explain the Sources of Public Revenue?	7	3	5	
	b).	Define Inflation and mention its causes?	7	3	5	
UNIT – V						
9.	a)	Define Money and write the functions of Money?	7	2	6	
	b)	What are the components of Money Supply?	7	2	6	
OR						
10.	a)	Explain the phases of a Trade Cycles?	7	2	6	
	b)	Explain the Cobweb theory of Trade Cycles?	7	2	6	